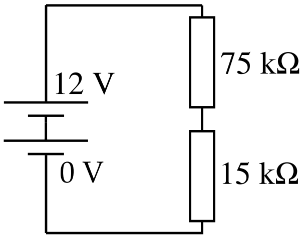
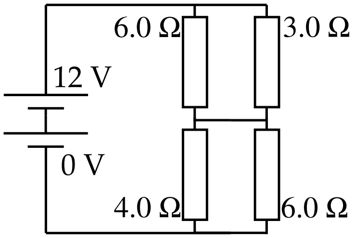
C5 Potential Dividers



(A)



(B)

C5.1 What is the potential difference across the lower resistor in circuit (A)?

C5.2 What is the potential difference across the $4.0\ \Omega$ resistor in circuit (B)?

C5.3 A light dependent resistor (LDR) is connected to a 5.0 V power supply in series with a $10\ \mathrm{k}\Omega$ resistor. For this LDR, its resistance is inversely proportional to the light level. So if the room gets twice as bright, the resistance will halve. During an experiment, the potential difference across the fixed resistor falls from 3.0 V to 1.0 V. Give the new level of light intensity as a fraction of the original light intensity to 2 significant figures.

Answers

C5.1 $12 \times \dfrac{15}{15+75}$ =2.0 V

C5.2 Top part has resistance 2.0 Ω. Bottom part has resistance 2.4 Ω.

Voltage across bottom part =12 V $\times \dfrac{2.4}{2.0+2.4}$ =6.5454 V

C5.3 Original resistance of LDR =10 k $\Omega \times \frac{2}{3} = \frac{20}{3}$ k Ω

New resistance of LDR =10 k $\Omega \times \frac{4}{1}$ =40 k Ω

New resistance as fraction of old = $\dfrac{40}{20/3} = 6$

So new light level is $\frac{1}{6}$ of original light level.